Durgasantosh G

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years in Java and Python, currently building AI agent workflows with Google ADK for Java on Citigroup's financial platforms (8 microservices, 500+ internal users). Built and run rejekt.ai, a live Generative AI interview platform using RAG, LangGraph, OpenAI/Gemini, and real-time Whisper voice.

SKILLS

Programming Languages: Python, Java 17, TypeScript, SQL

AI/ML: Generative AI, LLMs, RAG, AI agents, Prompt engineering, Vector embeddings, Model fine-tuning, LLM evaluation, Model Context Protocol (MCP), OpenAI API, Google Gemini, Google ADK

Frameworks & Libraries: LangChain, LangGraph, PyTorch, TensorFlow, SentenceTransformers, FastAPI, Spring Boot, React.js

Databases: PostgreSQL, Redis, DynamoDB, ChromaDB

Tools & Software: Docker, Kubernetes, GitHub Actions, Jenkins, pytest, JUnit, Git

Cloud Platforms: AWS (ECS Fargate, S3, RDS, CloudWatch)

Soft Skills: Cross-functional collaboration, Technical documentation, Agile/Scrum

WORK EXPERIENCE

Citigroup | Software Engineer | New Jersey Jun 2025 - Present

- Built AI agent workflows with Google ADK for Java that automate 12 multi-step financial processes; agents call enterprise APIs only through deterministic business-rule checks, keeping each decision predictable and auditable. Cut manual processing steps by 30%.
- Maintain 15 secure REST endpoints (Spring Security, JWT, centralized validation and exception handling) that serve agent-integrated workflows at 50K+ requests a day and 99.9% availability.
- Build and maintain end-to-end financial workflows in React.js, Java 17, and Spring Boot used by 500+ internal users across 8 production microservices.
- Work on Kafka and DynamoDB event pipelines that feed agent and UI services, validating payloads and tracing failed or delayed transactions back to their source.
- Cut page-load time from 3.2s to 2.4s by removing redundant API calls, memoizing expensive components, and simplifying state management.
- Investigated and resolved 20+ production incidents using application logs, API traces, event payloads, and Python scripts for log analysis, then added JUnit/Mockito regression tests so the same defects did not return.
- Shortened code-to-production time from 2 days to 6 hours by automating build and release steps across Docker, Kubernetes, Jenkins, and GitLab CI.
- Contributed to architecture reviews and API-contract decisions for 6 team-owned services, including the services agent workflows depend on.

Zensar Technologies | Software Developer | Hyderabad, India Jun 2022 - Jul 2023

- Developed and maintained full-stack features using Angular, React.js, Java, and Spring Boot for a microservices-based financial-services platform.
- Designed and enhanced RESTful APIs with request validation, exception handling, Spring Data JPA/Hibernate database interactions, and downstream service integration, maintaining stable API contracts across releases.
- Created reusable frontend components, shared validation utilities, and common data-handling workflows, reducing repetitive development effort by approximately 30%.
- Integrated frontend modules with backend services, handling asynchronous requests, loading states, validation failures, and API errors to provide reliable user experiences during degraded service conditions.
- Strengthened transaction-processing workflows through improved validation and exception handling, reducing processing errors by approximately 20% within supported financial modules.
- Updated Kafka producers and consumers, validated message payloads, and analyzed failed events to resolve transaction-processing issues in asynchronous workflows.

- Applied secure coding practices to PCI DSS–aligned modules by preventing sensitive payment information from appearing in application logs, exception messages, and API responses.
- Investigated cross-stack defects using application logs, API responses, and JUnit/Mockito regression tests, coordinating fixes with developers and QA engineers before release.
- Supported application builds and deployments using Jenkins, Docker, and AWS, monitoring releases for configuration and runtime issues within Agile delivery cycles.

Zensar Technologies | Junior Software Developer | Hyderabad, India Jun 2021 - May 2022

- Contributed to frontend development for a financial-services application using Angular and React.js, building reusable UI components, forms, and client-side validation.
- Integrated frontend modules with existing REST APIs, handling asynchronous requests, loading states, validation failures, and user-facing error messages.
- Reproduced reported issues by reviewing application logs and network responses, identified root causes, and implemented fixes with senior developer support.
- Created and maintained unit tests, addressed defects identified during regression testing, and participated in peer code reviews to improve application reliability.
- Managed assigned stories and defects using Git and Jira while participating in sprint planning, daily stand-ups, and other Agile development activities.

PROJECTS

rejekt.ai – Generative AI Interview Preparation Platform Jan 2026 - Present

- Designed and developed AI agents and agentic workflows for a Generative AI interview preparation platform that generates personalized technical, behavioral, and company-specific mock interviews using Large Language Models (LLMs).
- Built end-to-end Retrieval-Augmented Generation (RAG) pipelines leveraging SentenceTransformers, vector embeddings, and the ChromaDB vector database to deliver context-aware interview questions and feedback.
- Implemented prompt engineering and context engineering techniques to maintain conversational state and personalize multi-turn interview sessions.
- Built a LangGraph-based adaptive interview workflow that generates resume- and job-specific questions, evaluates voice/text responses for technical accuracy, STAR structure, filler words, and missing concepts, and dynamically triggers targeted follow-up questions.
- Integrated OpenAI and Google Gemini into application workflows for interview generation, answer evaluation, candidate scoring, and conversational AI, including tool/data integrations via the Model Context Protocol (MCP).
- Engineered real-time voice interview experiences using Speech-to-Text (Whisper), Text-to-Speech (TTS), and automated speech-processing pipelines.
- Developed LLM evaluation methodologies and AI-driven assessment systems that analyze communication skills, technical competency, behavioral responses, and interview readiness through LLM-based scoring frameworks.
- Built scalable backend services using Python, FastAPI, PostgreSQL, and Redis, deployed via AWS ECS Fargate, Docker, and GitHub Actions CI/CD with pytest-based automated testing.
- Developed large-scale data pipelines that collect, normalize, and analyze job listings across 19 sources, processing over 2.02 million listings and retaining approximately 115,600 qualified records.

Beyond Sight: Empowering Visually Impaired Users with Audible Graphs – UNT ACM W4A '24 Research Paper Sep 2023 - Apr 2024

- Co-developed a machine learning accessibility system supporting 3 chart formats (bar, line, and pie) that converts Excel charts and image-based graphs into audio narratives and structured text.
- Fine-tuned and benchmarked an ImageNet-pretrained ResNet deep learning model using PyTorch and TensorFlow, achieving 92% accuracy in chart classification and data extraction.
- Engineered an OCR + NLP pipeline (Pandas, NumPy, spaCy, Tesseract) applying data modeling and statistical analysis to extract titles, axes, legends, and values from data visualizations with 85% precision.
- Integrated Text-to-Speech (TTS) modules with screen-reader-compatible output, enabling visually impaired users to interpret chart information through real-time audio feedback.
- Co-authored the research paper presented at ACM W4A '24, contributing to accessible AI and digital inclusion research.

EDUCATION

University of North Texas Aug 2023 - May 2025
Master of Science, *Computer Science* GPA: 3.82

Vellore Institute of Technology University Aug 2018 - May 2022
Bachelor of Technology, *Computer Science*

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Developer – Associate - AWS